

# EEGMoE: A Domain-Decoupled Mixture-of-Experts Model for Self-Supervised EEG Representation Learning

Xuange Gao<sup>1</sup>, Graduate Student Member, IEEE, Danli Wang<sup>2</sup>, and Yanyan Zhao<sup>3</sup>

**Abstract**—Existing deep learning models for electroencephalogram (EEG) are typically tailored for specific tasks, datasets, or even subjects. This specialization restricts their applicability, reducing both perceptual capabilities and overall generalizability. Recent research has begun to explore large-scale pretraining for EEG representation learning. These efforts often focus on enabling data from different domains to be encoded by the same model through unifying the data format. However, this approach tends to overlook the importance of decoupling EEG representations across various domains, leading to the missing of valuable domain-specific details. Given the diverse distribution of large-scale EEG data forming multiple distinct domains, we hope to learn not only domain-shared representations but also to learn domain-specific representations. Therefore, in this article, we propose EEG mixture of experts (EEGMoE), a domain-decoupled self-supervised pretraining model for EEG representation learning. Specifically, EEGMoE introduces a Transformer-based domain-decoupled encoder, featuring specific and shared expert groups within our mixture-of-experts (MoE) block. The specific expert group adopts Top-K routing to select the most appropriate  $K$  experts for each token, while the shared expert group employs soft routing, leveraging all experts to learn for each token. Pretrained on various EEG datasets from multiple tasks, EEGMoE is then fine-tuned and validated on new datasets covering three mainstream EEG tasks, including emotion recognition (ER), motor imagery (MI) classification, and mental workload detection. Experimental results show that EEGMoE outperforms state-of-the-art models on three public datasets, demonstrating its strong generalization ability to new domains. Extensive experiments and visualizations further highlight the importance and effectiveness of disentangling domain-specific representations. Our code and model will be released.

**Index Terms**—Domain decouple, electroencephalogram (EEG) representation learning, mixture of experts (MoE), self-supervised learning.

## I. INTRODUCTION

**E**LECTROENCEPHALOGRAPH (EEG), a physiological signal that directly measures brain activities [1], [2], has proven to be indispensable in a range of neurological applications. These applications include, but are not limited

Received 13 December 2024; revised 7 July 2025 and 27 September 2025; accepted 3 January 2026. This work was supported in part by Beijing Natural Science Foundation under Grant L232098, in part by the Natural Science Foundation of Beijing and Key Project of Science and Technology Plan of Beijing Municipal Education Commission under Grant 21JD0044, and in part by the National Natural Science Foundation of China under Grant 61872363. (Corresponding author: Danli Wang.)

The authors are with the State Key Laboratory of Multimodal Artificial Intelligence Systems, Institute of Automation, Chinese Academy of Sciences, Beijing 100190, China, and also with the School of Artificial Intelligence, University of Chinese Academy of Sciences, Beijing 100049, China (e-mail: gaouxuange2022@ia.ac.cn; danli.wang@ia.ac.cn; zhaoyanyan2020@ia.ac.cn).

Digital Object Identifier 10.1109/TNNLS.2026.3652277

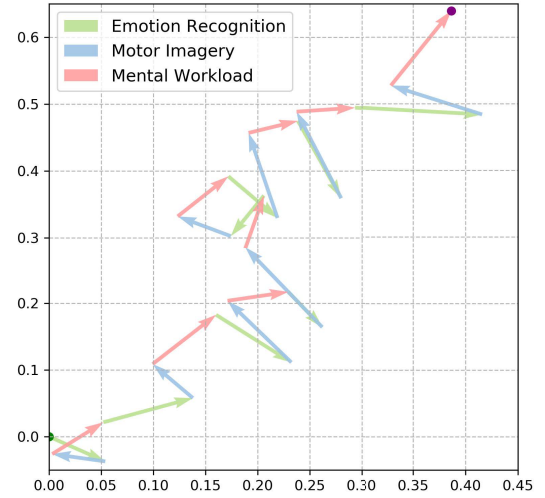


Fig. 1. Gradient optimization direction visualized in 2-D space of model parameters at each step during alternate self-supervised pretraining on emotional recognition task (DEAP [32]), MI classification task (BCI Competition IV-2a [33]), and mental workload detection task (STEW [34]) in a classical Transformer [35] model. As observed in this figure, during the gradient update process, the ER task drives the model parameters to optimize predominantly in the lower right direction, while the MI classification task guides the optimization toward the upper left direction. In contrast, the mental workload detection task steers the optimization toward the upper right direction.

to, emotion recognition (ER) [3], motor imagery (MI) classification [4], and mental workload detection [5].

Numerous models and handcrafted feature extraction methods have been proposed to boost the aforementioned fields. Nevertheless, these models or features are commonly tailored to particular tasks or domains, hindering their transferability and generalization at the task level. Specifically, some studies utilize differential entropy or power spectral density features combined with convolutional neural networks (CNNs) or recurrent neural networks (RNNs) to learn subject-dependent spatial-temporal-frequency information [6], [7], [8]. Other studies focus on analyzing brain connectivity and conducting subject-dependent mental workload detection [9], [10]. Additionally, some works leverage CNN, graph neural network (GNN), or Transformer backbones to directly model raw EEG signals under intersubject settings of some specific datasets or tasks [11], [12], [13], [14], [15], [16]. Although these model could adeptly handle the designated task, their task-specific training paradigm restricts its applicability to a single task, dataset, or even individual subject [15], [16], [17], [18], [19].

Another line of work pioneers the use of self-supervised pretraining to learn latent unified representations from extensive EEG data [20], [21], [22], [23]. These studies emphasize the uniformity of EEG input formats, indiscriminately processing all signals through the model without domain differentiation. However, leveraging homogeneous parameters to model heterogeneous data inevitably overlooks the potential discrepancies and conflicts inherent among different tasks as shown in Fig. 1. Different tasks lead to the model parameters being optimized in distinct directions during the learning process. In contrast, for the same task, the gradient update directions remain relatively stable across consecutive steps. This arises from the significant differences in EEG signals across subjects, datasets, and tasks, resulting in the formation of distinct domains within the EEG data distribution [18], [24], [25], [26]. This heterogeneity arises because EEG tasks inherently differ in their neural correlates, dominant frequency bands, and temporal dynamics [27], [28], [29], [30]. Such discrepancies naturally induce gradient conflicts during joint training, which explains the inconsistency in optimization directions [31]. The current undecoupled pretraining paradigm fails to fully unleash the potential of EEG models. Consequently, their performance does not exhibit a significant advantage over the task-specific models.

Recent advances in mixture-of-experts (MoE) architectures have shown great promise in handling heterogeneous data by conditionally activating a subset of specialized expert networks for different inputs. In contrast to traditional models that apply the same parameters to all inputs, MoE enables dynamic routing of data to task- or domain-specific experts, thereby improving both parameter efficiency and task adaptability. Such architectures have been successfully applied in natural language processing (NLP) [36], [37] and computer vision (CV) [38], [39]. In the EEG domain, a few works have used MoE-based architectures. However, most of them adopt a weighted-sum approach, aggregating the outputs of all experts rather than selectively activating them [40], [41], [42]. This fully shared structure undermines the core advantage of MoE, its ability to specialize and selectively route information. Other studies treat MoE as a multitask compatibility mechanism [43], which also falls short in capturing the intrinsic patterns underlying EEG signal heterogeneity across different tasks or domains. Inspired by the developments of MoE, we explore its potential in modeling heterogeneous EEG data across diverse tasks.

Therefore, to achieve robust EEG representation learning and ensure generalization across various domains, it is essential to consider both commonalities and specificities of each domain. In this article, we propose **EEGMoE**, a self-supervised pretraining model leveraging the MoE architecture. EEGMoE learns both domain-shared and domain-specific representations across diverse EEG data. Specifically, we propose the Transformer-based domain-decoupled encoder, which introduces a group of specific experts and shared experts within our **SSMoE** block to achieve the learning of the aforementioned representations. We adopt the Top-K routing strategy in the specific expert group to adaptively select the most appropriate experts for dealing with each token while

using the soft routing strategy in the shared expert group to fully utilize all experts for learning from every token. The former aims to flexibly learn fine-grained, domain-specific representations through various expert combinations. The latter learns from all domains and aims to acquire domain-shared representations from fixed experts. We pretrain EEGMoE across various EEG datasets of different tasks using masked signal reconstruction and fine-tune it on new EEG datasets. To evaluate the effectiveness of the proposed model, we conduct extensive experiments on three mainstream EEG tasks encompassing ER, MI classification, and mental workload detection. Our contribution is threefold.

- 1) We propose a novel self-supervised EEG representation learning model, EEGMoE, which is a unified model capable of effectively learning representations from unlabeled EEG data across various domains, thereby enhancing downstream tasks. Consequently, a single pretrained EEGMoE model can be easily transferred to various downstream datasets.
- 2) EEGMoE disentangles domain-specific representations and captures domain-shared representations through a specific and shared MoE block. To the best of our knowledge, this is the first work to explore the potential of learning decoupled EEG representations by introducing the concept of MoE.
- 3) Our EEGMoE achieves state-of-the-art performance across multiple benchmarks for ER, MI classification, and mental workload detection. Extensive experiments and visualizations further demonstrate its robust capability to learn both domain-shared and domain-specific EEG representations.

## II. RELATED WORKS

### A. EEG Representation Learning

Since the challenges posed by models tailored to a single subject or dataset and the successes achieved in learning generic representations in the fields of CV and NLP [44], [45], unified EEG representation learning models are being developed [21], [22], [23], [46]. These models merge information from multiple datasets into a single framework. MMM [21] proposed a pretraining framework with multidimensional position encoding, multilevel channel hierarchy, and multistage pretraining strategy to learn topology-agnostic EEG representations. LaBraM [22] adopted vector-quantized neural spectrum prediction to train a tokenizer that encodes EEG signal patches into compact codes. It aimed to overcome the montage differences between various EEG datasets and learn the unified representations. BIOT [46] proposed a flexible biosignal encoder that enables pretraining on multiple datasets by tokenizing signals of different formats into the same sentence structure. DMAE-EEG [23] leveraged neuroscientific priors to segment EEG into spatially and temporally coherent patches and employs a masked autoencoder to reconstruct clean signals from partially masked and noisy inputs. Some studies have also aimed at exploring more suitable self-supervised learning paradigms [47], [48]. However, current EEG representation learning mainly focuses on unifying vari-

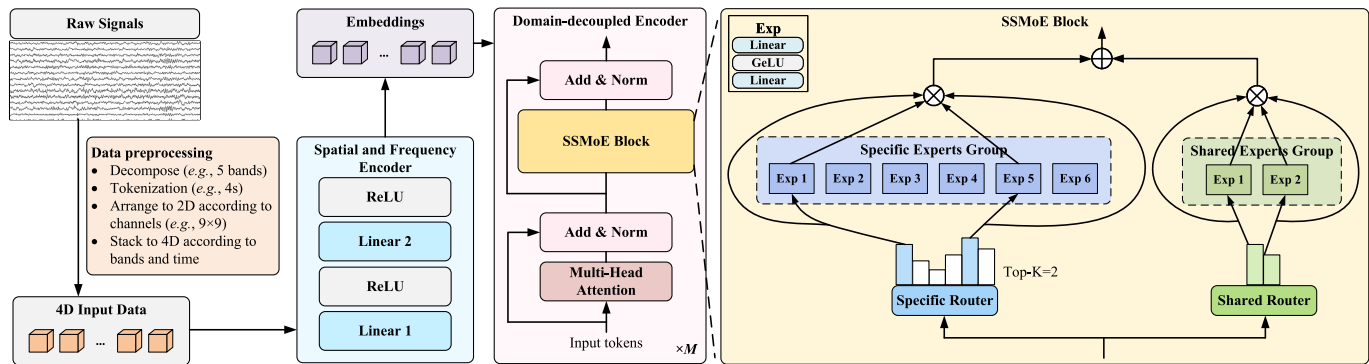


Fig. 2. Overall architecture of EEGMoE. After data preprocessing, as detailed in Section IV-A2, the raw EEG signals are converted into 4-D input data with the same format, where the four dimensions correspond to the temporal axis, frequency bands, and the 2-D spatial arrangement of electrodes. This 4-D input is then fed into spatial and frequency encoders to obtain meaningful embeddings. These embeddings are further processed by the domain-decoupled encoder, which includes multihead self-attention modules and the critical SSMoE block, to learn domain-shared and domain-specific representations.

ous signal formats but neglects the importance of decoupling EEG representations across different domains. As a result, they can only capture the commonalities and often fail to acquire the valuable specialities which are beneficial to the specific tasks.

Some studies in neuroscience and data analysis have indicated that EEG signals often exhibit significant differences across tasks, datasets, and even subjects [49]. Building upon the learning of domain-shared representations, further incorporating domain-specific representations can facilitate more comprehensive and meaningful representation learning from large-scale EEG data.

### B. Mixture of Experts

MoE [50] employs multiple expert modules that adaptively allow each expert to specialize in solving a specific subtask as determined by the router. The router determines which experts are employed for the given inputs. Previously, MoE has been used as a scaling tool in NLP [51] and CV [52]. Moreover, some works [38], [41] highlight the effectiveness and robustness of MoE in handling complex and heterogeneous data and adapting to various data distributions. In EEG classification, few studies have employed the concept of MoE. IDMMOE [40] proposed an identity-based multi-gate MoE for ER, which allocates part of model's subspace to each subject in a customized manner according to their identity. Seizure-MoE [41] adopted MoE structures for EEG-based seizure subtype classification. These works employ a weighted sum of all expert outputs at the level of input samples, which results in a lack of flexibility and efficiency. MoGE [42] introduced a mixture-of-graph-experts framework for cross-subject ER, where a router assigns EEG channels to specialized graph experts to decompose complex brain activity into simpler functional regions. However, it essentially remains a supervised feature extractor tailored for a single EEG task. EEGMamba [43] is a recent supervised model that incorporates an MoE architecture for multitask EEG classification within a bidirectional Mamba backbone. It introduces a task token that guides expert selection based on known task categories. However, MoE in EEGMamba primarily intended to support multitask compatibility, rather than to facilitate

the self-supervised learning of disentangled representations. Unlike multitask learning, which usually relies on labeled task categories and typically models differences at the level of separate task heads through supervised learning, our framework is designed for self-supervised domain-decoupled representation learning. Specifically, EEGMoE disentangles both domain-shared and domain-specific representations at a fine-grained feature level, enabling robust modeling of heterogeneous EEG data across tasks, datasets, and subjects during pretraining.

Moreover, to the best of our knowledge, our work takes a pioneering step in exploring the potential of MoE for learning both domain-shared and domain-specific representations from large-scale, diverse EEG data.

## III. METHODS

### A. Overview

The overall architecture of the EEGMoE model is shown in Fig. 2. All EEG raw signals are first preprocessed into a 4-D format. Then, a spatial and frequency encoder will be applied to the 4-D input data to extract spatial and frequency information and obtain the embeddings. Afterward, the embeddings are passed into the domain-decoupled encoder, which consists of Transformer-based multihead self-attention modules as well as specific and shared MoE blocks. Finally, the outputs of the domain-decoupled encoder are subsequently fed into the classifier to complete the downstream task. In the EEGMoE model, the spatial and frequency encoder, composed of a multilayer perceptron with two linear layers, primarily functions to generate embeddings. In contrast, the domain-decoupled encoder serves as the core component of the model, responsible for learning domain-shared and domain-specific representations. Section III-B will provide a detailed introduction to the domain-decoupled encoder. Additionally, we will also introduce our training framework in Section III-C.

### B. Domain-Decoupled Encoder

Our domain-decoupled encoder is based on the Transformer architecture and incorporates specific and shared MoE (SSMoE) block, as illustrated in Fig. 2. This block consists of two parts: the specific MoE and the shared MoE. The specific



MoE employs the Top-K routing strategy, primarily aimed at learning domain-specific representations, while the shared MoE utilizes the soft routing approach, focusing on learning domain-shared representations. By performing a weighted summation of the representations learned by the specific MoE and the shared MoE, a representation that integrates both domain generalization and decoupling is obtained.

1) *Specific MoE*: For the specific MoE, an MoE block substitutes the feedforward network (FFN) of the Transformer layer with an expert network comprising  $E$  experts  $\{e_i\}_{i=1}^{|E|}$ . Specifically, each expert consists of a two-layer multilayer perceptron (MLP) with a GELU activation function in between. For each input token  $x$ , the activation probabilities  $p_i(x)$  for each expert are determined by the router  $W_e$ , which is as follows:

$$g_x = W_e \cdot x \quad (1)$$

$$p_i(x) = \frac{\exp(g_{x_i})}{\sum_{j=1}^{|E|} \exp(g_{x_j})} \quad (2)$$

where  $g_x$  represents the routing scores assigned by the router to each expert for a given input token  $x$ . After calculating the probability  $p_i(x)$  of each expert, we used the Top-K routing strategy [53] to choose the most appropriate experts for each token. The weighted output of the specific MoE SpecMoE( $x$ ) can be formulated as follows:

$$\text{SpecMoE}(x) = \sum_{i \in \text{TopK}} p_i(x) e_i(x) \quad (3)$$

where  $e_i(x)$  is the output of the  $i_{\text{th}}$  expert that is an FFN.

2) *Shared MoE*: While the Top-K routing strategy can model the domain-specific representations of EEG data across various domains, the commonality of the EEG signals within different domains is still worth considering. Therefore, we propose the addition of a shared MoE, which employs the soft routing strategy with fixed experts to comprehensively learn from all data, thereby obtaining domain-shared representations. Specifically, the shared MoE is also based on the same multihead self-attention module and uses a fixed expert network comprising  $F$  experts  $\{f_i\}_{i=1}^{|F|}$ . For each input token  $x$ , the activation probabilities  $p_i(x)$  for each expert are also determined by the router  $W_f$  as (1) and (2). However, these  $F$  experts will be employed no matter which input token is. In other words, these experts will be used for learning the shared representation within all data of all domains. The weighted output ShareMoE( $x$ ) can be formulated as follows:

$$\text{ShareMoE}(x) = \sum_{i \in F} p_i(x) f_i(x) \quad (4)$$

where  $f_i(x)$  is the output of the  $i_{\text{th}}$  expert that is an FFN.

3) *Specific and Shared MoE*: Based on the specific MoE and shared MoE mentioned above, we further propose the SSMoE, which can comprehensively incorporate both speciality as well as commonality when determining the experts to be employed. Specifically, we combine the Top-K routing specific expert network with the soft routing shared expert network and form the SSMoE block, the formula is as follows:

$$\text{SSMoE}(x) = \text{SpecMoE}(x) + \text{ShareMoE}(x). \quad (5)$$

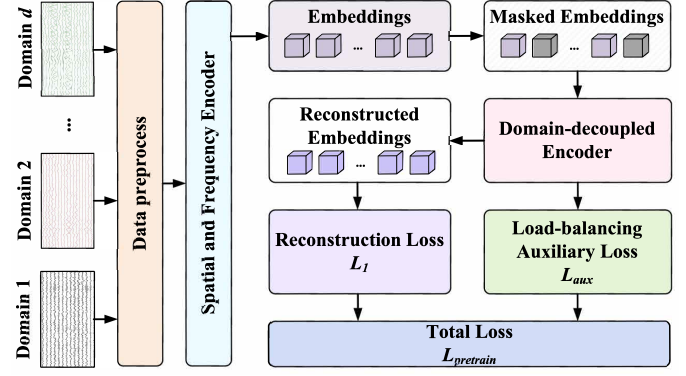


Fig. 3. Overview of the self-supervised pretraining stage of EEGMoE.

### C. Training Framework

The training framework of EEGMoE consists of two stages: self-supervised pretraining stage on large-scale EEG data for EEG representation learning and supervised downstream fine-tuning stage for adapting to new datasets.

1) *Stage 1 (Self-Supervised Pretraining)*: We can jointly pretrain the EEGMoE on multiple large-scale unlabeled datasets. The pretraining process is illustrated in Fig. 3. First, the input data  $X$  are passed through the spatial and frequency encoder to aggregate along the channel and frequency band dimensions to obtain the embeddings  $Z$ . These embeddings are then randomly masked, creating the input for the SSMoE. The SSMoE is designed to learn the latent representations of EEG signals by reconstructing these embeddings. During this process, we employ  $L_1$  loss as the loss function for embedding reconstruction. The  $L_1$  loss can be formulated as follows:

$$\mathcal{L}_1 = \frac{1}{S} \sum_{i=1}^S |Z_i - \tilde{Z}_i| \quad (6)$$

where  $S$  is the number of tokens and  $\tilde{Z}$  is the reconstructed embedding.

In addition, it is beneficial for the MoE block to have sparse activation of experts for the given token [54]. In practice, however, the Top-K routing method may result in the majority of the tokens being allocated to a few overburdened experts, leading to a load imbalance during training. Consequently, it causes some experts to be very busy while other experts remain underutilized and untrained. To deal with this problem, we introduce the load-balancing auxiliary loss [55]  $\mathcal{L}_{\text{aux}}$ , which can be formulated as follows:

$$\mathcal{L}_{\text{aux}} = E \sum_i^E h_i D_i \quad (7)$$

where  $h_i$  is the fraction of tokens allocated to expert  $e_i$ , which can be formulated as follows:

$$h_i = \frac{1}{S} \sum_{x \in \mathcal{B}} \mathbb{1}\{i \in \text{TopK}(d(x))\}. \quad (8)$$

$D_i$  is the fraction of the router probability allocated for expert  $e_i$ , indicating the probability fraction to expert  $e_i$  across all

TABLE I

STATISTICAL DATA OF THE NINE DATASETS USED IN THIS WORK ACROSS THREE TASKS, INCLUDING TOTAL TIME OF EACH DATASET, SAMPLING RATE (SR) OF EEG SIGNALS, NUMBER OF CHANNELS, TIME WINDOW LENGTH ( $T$ ), AND NUMBER OF SEGMENTED SAMPLES ("SUB" REPRESENTS THE SUBJECT). † DATASETS USED FOR FINE-TUNING AND VALIDATION, WITH OTHERS USED FOR PRETRAINING

| Task | Datasets                    | Time/hour | Sr    | Channels | T/second | # Sample (sub $\times$ segment) |
|------|-----------------------------|-----------|-------|----------|----------|---------------------------------|
| ER   | DREAMER [56]                | 22.89     | 128Hz | 14       | 2        | 42803 ( $23 \times 1861$ )      |
|      | SEED [57]                   | 45.00     | 200Hz | 62       | 2        | 76140 ( $15 \times 5076$ )      |
|      | MAHNOB-HCI [58]             | 12.21     | 256Hz | 32       | 2        | 19344 ( $24 \times 806$ )       |
|      | DEAP† [32]                  | 21.33     | 128Hz | 32       | 2        | 7680 ( $32 \times 240$ )        |
| MI   | EEGMMIDB [59]               | 47.23     | 160Hz | 64       | 4        | 18437 ( $103 \times 179$ )      |
|      | BCI Competition IV-1 [60]   | 8.21      | 100Hz | 59       | 4        | 1400 ( $7 \times 200$ )         |
|      | BCI Competition IV-2a† [33] | 5.76      | 250Hz | 22       | 4        | 5184 ( $9 \times 576$ )         |
| MWL  | EEGMat [61]                 | 6.00      | 500Hz | 23       | 1        | 8296 ( $34 \times 244$ )        |
|      | STEW† [34]                  | 3.75      | 128Hz | 14       | 5        | 2880 ( $48 \times 60$ )         |

tokens in the batch  $\mathcal{B}$ , which can be formulated as follows:

$$D_i = \frac{1}{S} \sum_{x \in \mathcal{B}} d_i(x) \quad (9)$$

where  $d_i(x)$  is the probability of routing token  $x$  to expert  $e_i$ .

Therefore, during the self-supervised pretraining stage, the total loss is as follows:

$$\mathcal{L}_{\text{pretrain}} = \mathcal{L}_1 + \alpha \mathcal{L}_{\text{aux}} \quad (10)$$

where  $\alpha$  is a hyperparameter to control the ratio between the two losses. We set  $\alpha$  to  $1 \times 10^{-4}$  in our work.

2) *Stage 2 (Supervised Downstream Fine-Tuning)*: Aiming to transfer knowledge among different datasets and gain improvements on the new task compared to training from scratch, we further propose the supervised downstream fine-tuning. Specifically, in stage 2, we inherit the parameter weights of the pretrained domain-decoupled encoder and connect a simple classifier of the specific downstream dataset after it. We then fine-tune the model through supervised learning with the labeled downstream dataset. We employ only a linear layer as the classifier, which is enough for the downstream task classification. We employ the cross-entropy loss  $\mathcal{L}_{\text{cls}}$  as the loss function, and it can be formulated as follows:

$$\mathcal{L}_{\text{cls}} = -\frac{1}{N} \sum_{i=1}^N \sum_{j=1}^C y_{ij} \log \hat{y}_{ij} \quad (11)$$

where  $N$  is the number of samples and  $C$  is the number of categories.

#### IV. EXPERIMENTS AND RESULTS

##### A. Experimental Setup

1) *Datasets*: We use the following nine datasets in the experiments, which can be divided into three categories based on their tasks. For ER, we use DREAMER [56], SEED [57], MAHNOB-HCI [58], and DEAP [32]. For MI classification, we use the EEG Motor Movement/Imagery dataset (EEGMMIDB) [59], BCI Competition IV-1 [60], and BCI Competition IV-2a [33]. For mental workload (MWL) detection, we use EEGMat [61] and STEW [34]. We use the DEAP, BCI

Competition IV-2a (BCIC4-2a), and STEW for fine-tuning and validation, while the other six datasets are used for pretraining. In detail, the DEAP dataset contains two subtasks for the ER of valence and arousal, both of which are two-category classifications. The BCIC4-2a dataset involves a four-class MI task, and the STEW encompasses a three-class mental workload task. The dataset statistics can be found in Table I, and the detailed descriptions of these datasets are provided as follows.

- 1) *DREAMER [56]*: An ER dataset based on the dimensional model, which contains 14 channels of EEG data at a 128-Hz sampling rate of 23 participants. Each participant viewed 18 film clips to evoke specific emotions. Each film lasted for a period of time between 65 and 393 s.
- 2) *SEED [57]*: An ER dataset based on the discrete model, which contains 62 channels of EEG data at a 200-Hz sampling rate of 15 participants. Each participant viewed 15 Chinese movie clips to evoke specific emotions. For each participant, there were 45 trials of EEG data in total, where each trial was approximately 4 min.
- 3) *MAHNOB-HCI [58]*: An ER dataset based on the dimensional model, which contains 32 channels of EEG data at a 256-Hz sampling rate of 27 participants. Each participant viewed 20 video clips to evoke specific emotions. Each clip lasted for a duration from 35 to 117 s.
- 4) *DEAP [32]*: An ER dataset based on the dimensional model, which contains 32 channels of EEG data at a 128-Hz sampling rate of 32 participants. Each participant viewed 40 1-min music videos to evoke specific emotions. The EEG data recorded for each trial (defined as a participant watching one video) includes 3-s baseline signals of relax and 60-s emotional signals.
- 5) *EEG Motor Movement/Imagery Dataset [59]*: An MI dataset, which contains 64 channels of EEG data at a 160-Hz sampling rate of 109 participants. Each participant performed 14 experimental runs: two 1-min baseline runs (one with eyes open and one with eyes closed) and three 2-min runs of each of the four tasks (two motor movement tasks and two MI tasks).

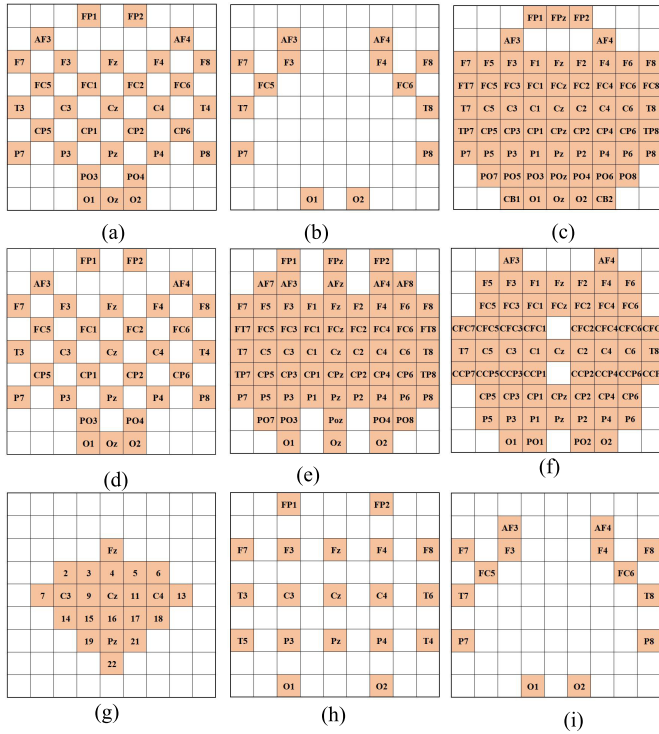


Fig. 4. Electrodes arrangement for 2-D brain maps of each dataset. (a) DEAP. (b) DREAMER. (c) SEED. (d) MAHNOB-HCI. (e) EEG Motor Movement/Imagery. (f) BCI Competition IV-1. (g) BCI Competition IV-2a. (h) EEGMAT. (i) STEW.

- 6) *BCI Competition IV-1* [60]: An MI dataset, which contains 59 channels of EEG data at a 100-Hz sampling rate of 7 participants. Each participant performed two classes of MI tasks, and each task contained 100 trials. Each trial lasted for 4 s.
- 7) *BCI Competition IV-2a* [33]: An MI dataset, which contains 22 channels of EEG data at a 250-Hz sampling rate of 9 participants. Each participant performed 2 sessions of experiments, each session contains six runs, and each run contains 48 trials (12 trials for each MI task). Therefore, there are 576 trials in total for each participant, and each trial lasted for 4 s.
- 8) *EEGMat* [61]: A mental workload dataset, which contains 23 channels of EEG data at a 500-Hz sampling rate of 36 participants. Each participant performed a mental arithmetic experiment that spanned a duration of 10 min and included three phases: adaptation (3 min), a resting state (3 min), and a mental arithmetic task-serial subtraction (4 min).
- 9) *STEW* [34]: A mental workload dataset, which contains 14 channels of EEG data at a 128-Hz sampling rate of 48 participants. Each participant performed a multitasking workload experiment that spanned a duration of 5 min and included two phases: a resting state (2.5 min) and a simultaneous capacity-based (SIMKAP) multitasking activity (2.5 min).

2) *Data Preprocess*: In order to incorporate temporal-spatial-frequency information within the input EEG data and reduce the differences resulting from various

TABLE II  
IMPLEMENTATION DETAILS

| Configuration            | Pre-train | Fine-tune                               |
|--------------------------|-----------|-----------------------------------------|
| Optimizer                | AdamW     |                                         |
| Warmup ratio             | 0.05      |                                         |
| Base LR                  | 1e-4      | 1e-3                                    |
| Batch size               | 64        | 128                                     |
| Load balance loss weight | 1e-4      | -                                       |
| Epoch                    | 5         | 35 (DEAP)<br>50 (BCIC4-2a)<br>15 (STEW) |

EEG montages, we adopt the 4-D input data preprocessing approach. Specifically, first, the raw EEG signals are divided into  $T_s$  long nonoverlapping segments. For each segment, the signals are decomposed into five frequency bands, including  $\delta(1 - 4\text{Hz})$ ,  $\theta(4 - 8\text{Hz})$ ,  $\alpha(8 - 14\text{Hz})$ ,  $\beta(14 - 31\text{Hz})$ , and  $\gamma(31 - 45\text{Hz})$ . For each band, the data are mapped into a 2-D recording to the locations of EEG electrodes on the scalp. Then, the 2-D brain maps are further smoothed using Gaussian filtering or  $z$ -score normalization to eliminate abrupt changes. Followed by this, the 2-D brain maps are stacked into 3-D along the frequency bands. Finally, the input data will be 4-D, for its format being  $X \in \mathbb{R}^{(T \times Sr) \times B \times H \times W}$ , where  $Sr$  and  $B$  represent the sampling rate of EEG signals and the number of frequency bands, respectively.  $H$  and  $W$  represent the height and width of the 2-D brain maps, respectively. The arrangement of EEG electrodes in the 2-D brain maps of each dataset is provided in Fig. 4.

3) *Implementation Details*: We conduct the leave-one-subject-out (LOSO) cross-validation on the DEAP, BCIC2-2a, and STEW datasets to verify the effectiveness of EEGMoE. For each dataset, we use the data of one subject as the test set and the data of the remaining subjects as the training set. Our model is implemented with PyTorch library in Python 3.8.0 with NVIDIA A800-SXM4-80G GPUs. For the STEW and BCIC4-2a datasets, where the class distributions are relatively balanced, we use accuracy (acc) and standard deviation (std) as evaluation metrics, following common practice in prior works. In contrast, since the DEAP dataset exhibits slight class imbalance, we additionally report AUC-PR and AUROC to provide a more comprehensive evaluation of model performance. Details on implementation and model hyperparameters are presented in Tables II and III, respectively.

4) *Baselines*: In order to demonstrate the effectiveness of EEGMoE and our self-supervised large-scale EEG data pretraining paradigm, we compare EEGMoE models fine-tuned from the same pretrained model against other current works. It is noteworthy that, in order to conduct a fair and comprehensive comparison, we not only compare EEGMoE with recent state-of-the-art models specifically designed for particular tasks but also replicate and compare it with several well-established models that have demonstrated robust performance and adaptability across various EEG recognition tasks. Details on the baselines that we replicate are as follows.



TABLE III  
HYPERPARAMETERS OF EEGMoE MODEL

| Spatial and Frequency Encoder |     |
|-------------------------------|-----|
| MLP hidden size               | 64  |
| MLP embed size                | 128 |
| Domain-decoupled Encoder      |     |
| Hidden size                   | 512 |
| Attention head                | 4   |
| Top-K                         | 2   |
| Shared experts                | 2   |
| Mask Ratio (pre-train)        | 0.4 |

- 1) *EEGNet* [66]: It is a compact convolutional neural network for EEG-based brain computer interfaces, which utilizes depthwise and separable convolutions for efficient feature extraction and classification.
- 2) *TSception* [11]: It is a multiscale convolutional neural network for EEG ER, which can learn discriminative representations in time and channel dimensions. It adopts a dynamic temporal layer to capture the dynamic temporal and frequency representations, while the asymmetric spatial layer is utilized to learn the discriminative global and hemisphere representations.
- 3) *EEG-Conformer* [67]: It is a compact convolutional Transformer model for EEG classification, which can encapsulate local and global features. It employs a convolution module to learn the low-level local features and utilizes a self-attention module to extract the global correlation within the local temporal features.
- 4) *LGGNet* [12]: It is a neurologically inspired graph neural network for EEG representation learning, which models the complex relations within and between functional regions of the brain.
- 5) *Seizure-MoE* [41]: Seizure-MoE adopts an MoE framework, in which preexisting deep neural network (DNN) models are used for feature extraction, and MLPs serve as the expert modules.
- 6) *BIOT* [46]: It is a self-supervised biosignal learning model that tokenizes biosignals of various formats into “sentences.” It employs a Transformer architecture to learn the representations of biosignals.
- 7) *LaBraM* [22]: It is a self-supervised EEG model designed for cross-dataset learning. It leverages vector-quantized neural spectrum prediction to train a neural tokenizer, which encodes EEG patches into compact neural codes.

### B. Comparison With Current Works

The results on the DEAP, BCIC4-2a, and STEW datasets are shown in Tables IV–VI, respectively, which indicate that EEGMoE outperforms the baselines across all datasets. Specifically, compared to the current state-of-the-art models, such as EEGFuseNet [62] for ER, DeepCNN [68] for MI, and the STEW model [34] for mental workload, our EEGMoE model demonstrates an accuracy improvement of 2.96%/4.18%, 6.01%, and 3.21% on the DEAP (valence/arousal), BCIC4-2a,

and STEW datasets, respectively. Even when compared with currently recognized influential EEG recognition models such as EEGNet [66], TSception [11], EEG-Conformer [67], and LGGNet [12], EEGMoE consistently achieves superior performance across three EEG downstream tasks. Models such as EEGNet and EEG-Conformer exhibit inconsistent and unstable performance across different tasks, with significant disparities in task-specific outcomes, highlighting their persistent limitations. In contrast, due to pretraining that learns domain-shared and domain-specific representations, EEGMoE demonstrates enhanced universality and generalization across various tasks. Compared to other self-supervised models, such as BIOT [46] and LaBraM [22], EEGMoE demonstrates significant performance superiority. This may be attributed to the fact that many of these self-supervised models primarily rely on a large amount of epilepsy data for pretraining representation learning. While they may capture certain patterns of EEG signals to some extent, the substantial differences between these pretraining tasks and downstream applications may hinder their ability to generalize effectively across a wide range of tasks.

### C. Ablation Studies

1) *Effect of Large-Scale Pretraining*: To evaluate the impact of pretraining EEGMoE on large-scale multidomain EEG data, we compare the performance of EEGMoE trained from scratch with that of the EEGMoE inheriting pretrained weights on three downstream tasks, as presented in Table VII. The results demonstrate that pretraining EEGMoE on large-scale EEG data to capture both domain-shared and domain-specific representations significantly enhances the performance on downstream tasks compared to training the model from scratch. This highlights the effectiveness of our pretraining stage, showing that learning shared and disentangled representations from extensive EEG data can benefit a range of downstream tasks.

2) *Effect of Specific and Shared Experts*: To validate the necessity of incorporating both flexibly selectable specific experts and shared experts in EEGMoE, we conduct ablation experiments on each component individually. Specifically, we set the number of shared experts to 4 and the Top-K for specific experts to 0, thereby removing the specific experts. For another configuration, we set Top-K to 4 and the number of shared experts to 0, removing the shared experts. Compared to the default configuration (Top-K of 2 and 2 shared experts), the former reduces the flexibility of the SSMoE block by impairing its ability to select expert combinations tailored to the current domain. In contrast, the latter prevents the model from comprehensively learning domain-shared representations from large-scale data. It is worth noting that, to minimize the influence of other factors, we kept the total number of experts fixed at 4 across the three models mentioned above, thereby avoiding any differences caused by variations in model parameter count. The results of the abovementioned models on the three datasets are presented in Table VIII. It is evident that the performance declines on all three datasets when either removing the specific experts or the shared experts. This demonstrates the effectiveness of both the specific and shared experts, highlighting their indispensability. Together,

TABLE IV

RESULTS OF DIFFERENT MODELS ON THE DEAP. DEAP-V AND DEAP-A CORRESPOND TO THE SUBTASKS OF RECOGNIZING VALENCE AND AROUSAL WITHIN THE DEAP DATASET. † RESULTS FROM OUR OWN REPRODUCTION

| Method               | DEAP-V                             |                                       |                                       | DEAP-A                              |                                       |                                       |
|----------------------|------------------------------------|---------------------------------------|---------------------------------------|-------------------------------------|---------------------------------------|---------------------------------------|
|                      | ACC $\pm$ STD(%)                   | AUC-PR $\pm$ STD                      | AUROC $\pm$ STD                       | ACC $\pm$ STD(%)                    | AUC-PR $\pm$ STD                      | AUROC $\pm$ STD                       |
| EEGFuseNet [62]      | 56.44                              | -                                     | -                                     | 58.55                               | -                                     | -                                     |
| TARDGCN [63]         | 57.73                              | -                                     | -                                     | 58.35                               | -                                     | -                                     |
| TAS-Net [64]         | 57.84                              | -                                     | -                                     | 60.51                               | -                                     | -                                     |
| RHPRNet [65]         | 58.06                              | -                                     | -                                     | 58.19                               | -                                     | -                                     |
| EEGNet† [66]         | 57.18 $\pm$ 7.09                   | 0.5642 $\pm$ 0.1190                   | 0.5086 $\pm$ 0.0920                   | 61.33 $\pm$ 12.24                   | 0.6020 $\pm$ 0.1712                   | 0.5360 $\pm$ 0.0652                   |
| TSception† [11]      | 55.90 $\pm$ 9.09                   | 0.5863 $\pm$ 0.1056                   | 0.5411 $\pm$ 0.0610                   | 61.41 $\pm$ 13.38                   | <b>0.6279 <math>\pm</math> 0.1594</b> | 0.5624 $\pm$ 0.0610                   |
| EEG-Conformer† [67]  | 55.77 $\pm$ 8.11                   | 0.6023 $\pm$ 0.1051                   | 0.5570 $\pm$ 0.0631                   | 59.68 $\pm$ 13.71                   | 0.6237 $\pm$ 0.1461                   | 0.5521 $\pm$ 0.0809                   |
| LGGNet† [12]         | 57.35 $\pm$ 5.80                   | <b>0.6183 <math>\pm</math> 0.0931</b> | <b>0.5633 <math>\pm</math> 0.0601</b> | 58.14 $\pm$ 10.75                   | 0.6062 $\pm$ 0.1717                   | 0.5264 $\pm$ 0.0820                   |
| Seizure-MoE† [41]    | 56.94 $\pm$ 6.09                   | 0.5810 $\pm$ 0.1115                   | 0.5234 $\pm$ 0.0599                   | 59.63 $\pm$ 12.32                   | 0.5980 $\pm$ 0.1508                   | 0.5302 $\pm$ 0.0721                   |
| BIOT† [46]           | 53.35 $\pm$ 5.76                   | 0.5920 $\pm$ 0.1026                   | 0.5301 $\pm$ 0.0768                   | 60.88 $\pm$ 11.93                   | 0.6213 $\pm$ 0.1627                   | 0.5629 $\pm$ 0.0605                   |
| LaBraM† [22]         | 56.83 $\pm$ 7.89                   | 0.5851 $\pm$ 0.0969                   | 0.5426 $\pm$ 0.0580                   | 58.72 $\pm$ 14.20                   | 0.5955 $\pm$ 0.1539                   | 0.5261 $\pm$ 0.0444                   |
| <b>EEGMoE (ours)</b> | <b>59.40 <math>\pm</math> 7.10</b> | 0.6171 $\pm$ 0.1598                   | 0.5517 $\pm$ 0.0719                   | <b>62.73 <math>\pm</math> 11.39</b> | 0.6245 $\pm$ 0.1626                   | <b>0.5693 <math>\pm</math> 0.0761</b> |

TABLE V

RESULTS [ACC  $\pm$  STD (%)] OF DIFFERENT MODELS ON THE BCIC4-2A. † RESULTS FROM OUR OWN REPRODUCTION

| Method               | BCIC4-2a                            |
|----------------------|-------------------------------------|
| DeepCNN [68]         | 41.91 $\pm$ 6.90                    |
| MCNN [69]            | 44.79 $\pm$ 5.76                    |
| SWPC [70]            | 45.04 $\pm$ 9.40                    |
| EEGNet† [66]         | 45.66 $\pm$ 11.86                   |
| TSception† [11]      | 43.68 $\pm$ 8.13                    |
| EEG-Conformer† [67]  | 30.63 $\pm$ 3.23                    |
| LGGNet† [12]         | 39.58 $\pm$ 4.96                    |
| SeizureMoE† [41]     | 46.23 $\pm$ 8.02                    |
| BIOT† [46]           | 41.10 $\pm$ 11.63                   |
| LaBraM† [22]         | 29.59 $\pm$ 4.29                    |
| <b>EEGMoE (ours)</b> | <b>47.92 <math>\pm</math> 11.53</b> |

TABLE VI

RESULTS [ACC  $\pm$  STD (%)] OF DIFFERENT MODELS ON THE STEW. † RESULTS FROM OUR OWN REPRODUCTION

| Method               | STEW                                |
|----------------------|-------------------------------------|
| STEW [34]            | 69.20                               |
| EEGNet† [66]         | 61.15 $\pm$ 19.15                   |
| TSception† [11]      | 68.41 $\pm$ 21.83                   |
| EEG-Conformer† [67]  | 68.63 $\pm$ 20.53                   |
| LGGNet† [12]         | 71.85 $\pm$ 20.78                   |
| SeizureMoE† [41]     | 57.04 $\pm$ 20.89                   |
| BIOT† [46]           | 69.70 $\pm$ 16.55                   |
| LaBraM† [22]         | 52.49 $\pm$ 12.35                   |
| <b>EEGMoE (ours)</b> | <b>72.41 <math>\pm</math> 18.99</b> |

they enable EEGMoE to simultaneously learn domain-specific and domain-shared representations.

3) *Effect of SSMoE Architecture*: To demonstrate the superiority of the SSMoE architecture employed by EEGMoE, which includes both shared and specific experts, we compare it

with a non-SSMoE architecture of equivalent model parameter count. Since the MoE model activates a different number of parameters during the training and inference phases, we define two comparison models: the activated equivalent model and the total equivalent model. The activated equivalent model has the same number of parameters activated during inference as EEGMoE, while the total equivalent model has the same total parameter count as the number of parameters activated during the training phase of EEGMoE. The results of the abovementioned models on the three datasets are presented in Table IX. The results indicate that, despite having comparable or even lower computational load than both the activated equivalent model and the total equivalent model, EEGMoE achieves superior performance. This underscores the importance of domain decoupling, a feature absent in non-SSMoE models. Additionally, this highlights the computational efficiency advantage of EEGMoE.

#### D. Parameter Analysis

1) *Effect of Number of Experts*: To better understand how the number of specific and shared experts affects the performance of our EEGMoE model, we conduct a series of experiments by varying the number of experts while keeping the overall model structure and training strategy consistent. Given the combinatorial explosion of possible setups, we select several commonly used MoE configurations for comparison. The results are shown in Table X. It can be observed that the number of experts should be chosen with moderation. When the number of specific expert candidates or the Top- $K$  value is either excessively small or large relative to the number of tasks, the advantages of the model cannot be fully realized. Moreover, the inclusion of more than one shared expert often leads to improved performance. These experimental results also indirectly demonstrate the rationality of the model design. From our analysis, we can also provide some empirical tips for practical configuration. For Top- $K$  routing, a small Top- $K$  (e.g., 2) is sufficient, which allows each token to combine multiple domain-specific experts without excessive redundancy. For shared experts, including more than



TABLE VII  
RESULTS [ACC  $\pm$  STD (%)] OF ABLATION STUDIES OF THE LARGE-SCALE PRETRAINING

| Model     | Pre-training | DEAP-V                             | DEAP-A                              | BCIC4-2a                            | STEW                                | Average      |
|-----------|--------------|------------------------------------|-------------------------------------|-------------------------------------|-------------------------------------|--------------|
| Scratch   | $\times$     | 58.62 $\pm$ 8.02                   | 61.86 $\pm$ 11.80                   | 45.98 $\pm$ 9.86                    | 70.52 $\pm$ 19.28                   | 59.25        |
| Fine-tune | $\checkmark$ | <b>59.40 <math>\pm</math> 7.10</b> | <b>62.73 <math>\pm</math> 11.39</b> | <b>47.92 <math>\pm</math> 11.53</b> | <b>72.41 <math>\pm</math> 18.99</b> | <b>60.62</b> |

TABLE VIII  
RESULTS [ACC  $\pm$  STD (%)] OF ABLATION STUDIES OF THE SPECIFIC EXPERTS AND SHARED EXPERTS WITHIN THE EEGMoE

| Specific experts | Shared experts | DEAP-V                             | DEAP-A                              | BCIC4-2a                            | STEW                                | Average      |
|------------------|----------------|------------------------------------|-------------------------------------|-------------------------------------|-------------------------------------|--------------|
| 0                | 4              | 58.89 $\pm$ 7.86                   | 62.32 $\pm$ 11.30                   | 45.21 $\pm$ 9.72                    | <b>72.48 <math>\pm</math> 19.42</b> | 59.73        |
| 4                | 0              | 58.46 $\pm$ 8.18                   | 61.12 $\pm$ 13.30                   | 42.16 $\pm$ 6.49                    | 70.96 $\pm$ 19.34                   | 58.18        |
| 2                | 2              | <b>59.40 <math>\pm</math> 7.10</b> | <b>62.73 <math>\pm</math> 11.39</b> | <b>47.92 <math>\pm</math> 11.53</b> | 72.41 $\pm$ 18.99                   | <b>60.62</b> |

TABLE IX  
RESULTS [ACC  $\pm$  STD (%)] OF ABLATION STUDIES OF THE SSMoE ARCHITECTURE

| Model                | # Total Params | # Activated Params | DEAP-V                             | DEAP-A                              | BCIC4-2a                            | STEW                                | Average      |
|----------------------|----------------|--------------------|------------------------------------|-------------------------------------|-------------------------------------|-------------------------------------|--------------|
| Activated Equivalent | 0.89M          | 0.89M              | 58.89 $\pm$ 7.86                   | 62.32 $\pm$ 11.30                   | 45.21 $\pm$ 9.72                    | <b>72.48 <math>\pm</math> 19.42</b> | 59.73        |
| Total Equivalent     | 1.68M          | 1.68M              | 58.16 $\pm$ 7.62                   | 62.30 $\pm$ 10.74                   | 42.93 $\pm$ 8.54                    | 72.26 $\pm$ 19.06                   | 58.91        |
| EEGMoE               | <b>1.68M</b>   | <b>0.89M</b>       | <b>59.40 <math>\pm</math> 7.10</b> | <b>62.73 <math>\pm</math> 11.39</b> | <b>47.92 <math>\pm</math> 11.53</b> | 72.41 $\pm$ 18.99                   | <b>60.62</b> |

TABLE X  
RESULTS [ACC  $\pm$  STD (%)] OF PARAMETER ANALYSIS OF THE EXPERTS NUMBER WITHIN THE EEGMoE

| Specific expert candidates | Top-K | Shared experts | DEAP-V                             | DEAP-A                              | BCIC4-2a                            | STEW                                | Average      |
|----------------------------|-------|----------------|------------------------------------|-------------------------------------|-------------------------------------|-------------------------------------|--------------|
| 4                          | 2     | 1              | 58.15 $\pm$ 7.92                   | 62.50 $\pm$ 11.49                   | 41.20 $\pm$ 6.67                    | 73.30 $\pm$ 19.40                   | 58.79        |
|                            |       | 2              | <b>59.65 <math>\pm</math> 7.69</b> | 62.86 $\pm$ 10.50                   | 40.65 $\pm$ 5.98                    | 74.07 $\pm$ 18.85                   | 59.31        |
| 6                          | 2     | 1              | 58.58 $\pm$ 7.34                   | 61.77 $\pm$ 12.33                   | 43.97 $\pm$ 8.60                    | 71.81 $\pm$ 19.03                   | 59.03        |
|                            |       | 2              | 59.40 $\pm$ 7.10                   | 62.73 $\pm$ 11.39                   | <b>47.92 <math>\pm</math> 11.53</b> | 72.41 $\pm$ 18.99                   | <b>60.62</b> |
| 8                          | 4     | 2              | 59.13 $\pm$ 7.64                   | 61.91 $\pm$ 12.64                   | 46.28 $\pm$ 10.25                   | <b>75.07 <math>\pm</math> 18.91</b> | 60.60        |
|                            |       | 2              | 58.97 $\pm$ 7.24                   | 63.35 $\pm$ 10.14                   | 41.42 $\pm$ 6.32                    | 71.93 $\pm$ 19.89                   | 58.92        |
|                            | 4     | 2              | 59.31 $\pm$ 7.47                   | 63.27 $\pm$ 10.85                   | 38.91 $\pm$ 4.67                    | 73.53 $\pm$ 18.30                   | 58.76        |
|                            |       | 2              | 59.11 $\pm$ 6.86                   | <b>63.40 <math>\pm</math> 10.63</b> | 38.49 $\pm$ 4.79                    | 72.96 $\pm$ 19.09                   | 58.49        |
|                            |       | 4              | 58.23 $\pm$ 8.16                   | 60.90 $\pm$ 13.01                   | 45.78 $\pm$ 8.57                    | 72.11 $\pm$ 21.23                   | 59.26        |

TABLE XI  
RESULTS [ACC  $\pm$  STD (%)] OF USING DIFFERENT OBJECTIVES DURING THE SELF-SUPERVISED PRETRAINING STAGE

| Objective | DEAP-V                             | DEAP-A                              | BCIC4-2a                            | STEW                                | Average      |
|-----------|------------------------------------|-------------------------------------|-------------------------------------|-------------------------------------|--------------|
| $L_1$     | <b>59.40 <math>\pm</math> 7.10</b> | <b>62.73 <math>\pm</math> 11.39</b> | <b>47.92 <math>\pm</math> 11.53</b> | 72.41 $\pm$ 18.99                   | <b>60.62</b> |
| $L_2$     | 59.04 $\pm$ 7.06                   | 60.22 $\pm$ 13.28                   | 45.24 $\pm$ 8.47                    | <b>74.15 <math>\pm</math> 18.04</b> | 59.66        |
| Cosine    | 58.67 $\pm$ 7.31                   | 62.20 $\pm$ 10.96                   | 39.51 $\pm$ 6.27                    | 72.93 $\pm$ 18.13                   | 58.33        |
| RP        | 58.74 $\pm$ 7.62                   | 62.62 $\pm$ 12.50                   | 46.16 $\pm$ 9.76                    | 72.89 $\pm$ 19.72                   | 60.10        |
| TS        | 57.99 $\pm$ 8.06                   | 61.94 $\pm$ 12.21                   | 44.35 $\pm$ 8.01                    | 73.04 $\pm$ 20.01                   | 59.33        |

one consistently improves generalization across tasks, as this ensures that commonalities are not underrepresented.

2) *Effect of Pretraining Objectives*: Commonly used loss functions for self-supervised reconstruction tasks include  $L_1$  loss,  $L_2$  loss, and Cosine loss. To identify the most suitable

loss function for EEG self-supervised tasks, we first evaluate the impact of different reconstruction losses on downstream performance. In addition to reconstruction losses, we further examine other pretraining objectives such as contrastive learning and temporal prediction. Specifically, inspired by [71], we

TABLE XII  
RESULTS [ACC  $\pm$  STD (%)] OF DIFFERENT FUSION MECHANISMS BETWEEN SPECIFIC AND SHARED EXPERTS

| Method                           | DEAP-V                             | DEAP-A                              | BCIC4-2a                            | STEW                                | Average      |
|----------------------------------|------------------------------------|-------------------------------------|-------------------------------------|-------------------------------------|--------------|
| Gated Fusion                     | 59.28 $\pm$ 6.89                   | 61.91 $\pm$ 13.04                   | 37.18 $\pm$ 4.71                    | 67.48 $\pm$ 16.79                   | 56.46        |
| Attention-based Fusion           | 59.18 $\pm$ 6.76                   | 62.42 $\pm$ 11.58                   | 36.16 $\pm$ 3.43                    | 70.85 $\pm$ 18.08                   | 57.15        |
| <b>Additive Fusion (default)</b> | <b>59.40 <math>\pm</math> 7.10</b> | <b>62.73 <math>\pm</math> 11.39</b> | <b>47.92 <math>\pm</math> 11.53</b> | <b>72.41 <math>\pm</math> 18.99</b> | <b>60.62</b> |

TABLE XIII  
RESULTS [ACC  $\pm$  STD (%)] OF USING DIFFERENT MASK RATIOS DURING THE SELF-SUPERVISED PRETRAINING STAGE

| Ratio      | DEAP-V                             | DEAP-A                              | BCIC4-2a                            | STEW                                | Average      |
|------------|------------------------------------|-------------------------------------|-------------------------------------|-------------------------------------|--------------|
| 0.2        | 58.16 $\pm$ 8.14                   | 61.37 $\pm$ 11.82                   | 46.55 $\pm$ 10.43                   | 70.93 $\pm$ 18.31                   | 59.25        |
| <b>0.4</b> | <b>59.40 <math>\pm</math> 7.10</b> | <b>62.73 <math>\pm</math> 11.39</b> | <b>47.92 <math>\pm</math> 11.53</b> | <b>72.41 <math>\pm</math> 18.99</b> | <b>60.62</b> |
| 0.6        | 58.53 $\pm$ 7.93                   | 61.87 $\pm$ 12.18                   | 47.09 $\pm$ 10.87                   | 70.15 $\pm$ 19.71                   | 59.41        |
| 0.8        | 58.76 $\pm$ 7.13                   | 61.09 $\pm$ 12.66                   | 45.93 $\pm$ 10.26                   | 72.33 $\pm$ 19.54                   | 59.53        |

adopt a contrastive loss based on relative positioning (RP) and a temporal prediction loss based on temporal shuffling (TS). The results are reported in Table XI. They show that the performance differences across downstream tasks are relatively small under different pretraining objectives, highlighting both the flexibility of our architecture to adapt to diverse pretraining paradigms and its robustness in effectively learning EEG representations. Based on these findings, we ultimately select the simplest and most effective  $L_1$  loss as the default objective.

3) *Effect of Expert Fusion Strategies*: To investigate the effect of different fusion strategies between the shared and specific expert groups, we compared the current additive fusion with two alternative strategies: gated fusion and attention-based fusion. For gated fusion, we introduce a gating network [72] to compute the relative weights between the shared and specific expert groups and then perform a weighted summation for fusion. For attention-based fusion, we concatenate the outputs of the shared and specific expert groups, feed them into a Transformer layer, and extract the class token to produce the fused representation. The results are presented in Table XII, which indicates that additive fusion consistently yields the best performance. The performance differences between the fusion methods suggest that additive fusion, which simply sums the outputs of the shared and specific experts, is sufficient for effectively balancing domain-specific and domain-shared representations in our model.

4) *Effect of Mask Ratio*: To explore the impact of the mask ratio of the embeddings during the pretraining stage, we pretrain the EEGMoE models with varying mask ratios and subsequently conduct experimental evaluations on downstream tasks. The results are shown in Table XIII, where we observe that a mask ratio of 0.4 yields the best performance across all three datasets. Additionally, it is evident that as the mask ratio increases from 0.2 to 0.8, the model's performance on downstream tasks first improves and then declines, indicating that both lower and higher mask ratios are detrimental to downstream task performance. This phenomenon may be attributed to the fact that a lower mask ratio leads to insufficient training during the pretraining stage, resulting

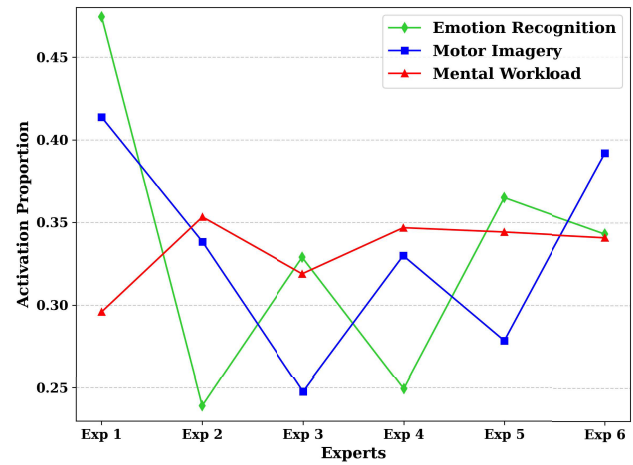


Fig. 5. Activation proportions of each expert within the specific expert group on three tasks.

in inadequate representation learning. Conversely, a higher mask ratio makes the reconstruction task more challenging, impeding the model's training process. Therefore, selecting an appropriate mask ratio is crucial for effective representation learning during the pretraining stage.

#### E. Visualization and Analysis

To further validate the effectiveness of the EEGMoE model in decoupling data across domains during the representation learning on large-scale EEG data, we conduct statistical and visual analysis of the specific experts' activation distributions across various tasks during the pretraining stage.

1) *Different Tasks Display Distinct Preferences for Experts*: We analyze the activation proportions of each expert within the specific expert group on ER, MI classification, and mental workload detection. As illustrated in Fig. 5, the horizontal axis represents different experts, while the vertical axis indicates the proportions of selections for each expert. It can be observed that ER exhibits a higher activation proportion for Experts 1,

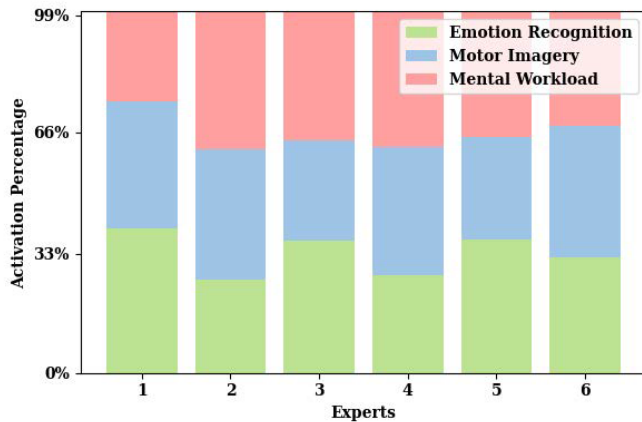


Fig. 6. Activation percentages of data handled by each expert within the specific expert group on the three tasks.

3, and 5, suggesting that this task tends to favor these experts for representation learning. In contrast, the MI classification shows a higher activation proportion for Experts 1, 4, and 6, with a notably lower activation for Experts 3 and 5. For mental workload detection, there are higher activation proportion for Expert 2 and lower activation proportion for Expert 1, in stark contrast to the pattern observed from the other two tasks. These findings indicate that these three tasks display distinct preferences for different experts, and the varying activation proportions across experts for different tasks further confirm that our pretraining stage successfully achieved decoupled learning for different tasks through the expert allocation.

2) *Different Experts Responsible for Distinct Tasks:* We analyze the percentage of data handled by each expert on the three tasks. As illustrated in Fig. 6, the horizontal axis represents different experts, while the vertical axis indicates the percentages of each expert handling the three tasks. The figure shows that Experts 1, 3, and 5 are primarily responsible for the ER, while Experts 2 and 4 mainly handle the mental workload detection. Expert 6 is primarily used for the MI classification. This specialization of different experts in distinct tasks further demonstrates that EEGMoE effectively decouples and learns representations for each specific task.

In summary, Fig. 5 answers: which experts does each task prefer? Fig. 6 answers: which task is each expert mainly responsible for? Figs. 5 and 6 jointly demonstrate how EEGMoE effectively decouples representations across tasks through expert specialization. Moreover, the distinct activation patterns observed across tasks further emphasize the necessity of such decoupling, as they reveal intrinsic differences in the way various tasks engage the model's experts.

## V. CONCLUSION

In this article, we propose EEGMoE, a domain-decoupled self-supervised pretraining model that leverages large-scale EEG data to learn both domain-shared and domain-specific representations, enhancing performance on multiple tasks. EEGMoE introduces a Transformer-based domain-decoupled encoder, allowing it to learn these representations across

various EEG domains. EEGMoE demonstrates the optimal performance on ER, MI classification, and mental workload detection. Ablation studies and visualizations further substantiate the efficacy of EEGMoE in decoupling large-scale EEG data across multiple domains. We hope that EEGMoE will offer a novel perspective and paradigm for EEG representation learning. In the future, we plan to incorporate a greater variety of tasks into EEGMoE, further expanding its applicability across diverse research domains and practical scenarios. Possible directions may include areas such as neuromodulation, brain-controlled robotics, fatigue detection, and clinical diagnostics, where generalizable EEG representations may offer promising opportunities. We hope that our work can serve as a foundation for further exploration in these and other emerging EEG-based applications.

## ACKNOWLEDGMENT

The authors would like to thank Tongtian Yue for his inspiration for the idea of this article, engaging discussions, meticulous review of the writing, timely assistance with the experiments, and numerous selfless contributions throughout this article.

## REFERENCES

- [1] A. Biasucci, B. Franceschiello, and M. M. Murray, "Electroencephalography," *Current Biol.*, vol. 29, no. 3, pp. R80–R85, 2019.
- [2] X. Wang, D. Wang, X. Gao, Y. Zhao, and S. C. Chiu, "Enhancing EEG-based decision-making performance prediction by maximizing mutual information between emotion and decision-relevant features," *IEEE Trans. Affect. Comput.*, vol. 15, no. 3, pp. 1228–1240, Jul. 2024.
- [3] Z. Zhang, S. Zhong, and Y. Liu, "Beyond mimicking under-represented emotions: Deep data augmentation with emotional subspace constraints for EEG-based emotion recognition," in *Proc. AAAI Conf. Artif. Intell.*, 2024, vol. 38, no. 9, pp. 10252–10260.
- [4] S. An, S. Kim, P. Chikontwe, and S. H. Park, "Dual attention relation network with fine-tuning for few-shot EEG motor imagery classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 11, pp. 15479–15493, Nov. 2024.
- [5] Z. Wang, Y. Ouyang, and H. Zeng, "ARFN: An attention-based recurrent fuzzy network for EEG mental workload assessment," *IEEE Trans. Instrum. Meas.*, vol. 73, pp. 1–14, 2024.
- [6] F. Shen, G. Dai, G. Lin, J. Zhang, W. Kong, and H. Zeng, "EEG-based emotion recognition using 4D convolutional recurrent neural network," *Cognit. Neurodynamics*, vol. 14, no. 6, pp. 815–828, Dec. 2020.
- [7] G. Xiao, M. Shi, M. Ye, B. Xu, Z. Chen, and Q. Ren, "4D attention-based neural network for EEG emotion recognition," *Cognit. Neurodynamics*, vol. 16, no. 4, pp. 1–14, Aug. 2022.
- [8] X. Gao, D. Wang, Y. Zhao, and X. Wang, "CFBM: Cubic filtered brain map creation method and models for EEG-based emotion recognition," in *Proc. IEEE 22nd Int. Symp. Biomed. Imag. (ISBI)*, Apr. 2025, pp. 1–5.
- [9] Y. Yan et al., "Topological EEG-based functional connectivity analysis for mental workload state recognition," *IEEE Trans. Instrum. Meas.*, vol. 72, pp. 1–14, 2023.
- [10] M. Safari, R. Shalhaf, S. Bagherzadeh, and A. Shalhaf, "Classification of mental workload using brain connectivity and machine learning on electroencephalogram data," *Sci. Rep.*, vol. 14, no. 1, p. 9153, Apr. 2024.
- [11] Y. Ding, N. Robinson, S. Zhang, Q. Zeng, and C. Guan, "TSception: Capturing temporal dynamics and spatial asymmetry from EEG for emotion recognition," *IEEE Trans. Affect. Comput.*, vol. 14, no. 3, pp. 2238–2250, Jul. 2023.
- [12] Y. Ding, N. Robinson, C. Tong, Q. Zeng, and C. Guan, "LGGNet: Learning from local-global-graph representations for brain-computer interface," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 7, pp. 9773–9786, Jul. 2024.
- [13] G. Siddhad, A. Gupta, D. P. Dogra, and P. P. Roy, "Efficacy of transformer networks for classification of EEG data," *Biomed. Signal Process. Control*, vol. 87, Jan. 2024, Art. no. 105488.



- [14] H. W. Aung, J. J. Li, Y. An, and S. W. Su, "A real-time framework for EEG signal decoding with graph neural networks and reinforcement learning," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 36, no. 9, pp. 17047–17058, Sep. 2025.
- [15] C. Li et al., "An efficient graph learning system for emotion recognition inspired by the cognitive prior graph of EEG brain network," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 36, no. 4, pp. 7130–7144, Apr. 2025.
- [16] C. Cheng, Z. Yu, Y. Zhang, and L. Feng, "Hybrid network using dynamic graph convolution and temporal self-attention for EEG-based emotion recognition," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 12, pp. 18565–18575, Dec. 2024.
- [17] S. Shao, G. Han, T. Wang, C. Lin, C. Song, and C. Yao, "EEG-based mental workload classification method based on hybrid deep learning model under IoT," *IEEE J. Biomed. Health Informat.*, vol. 28, no. 5, pp. 2536–2546, May 2024.
- [18] Y. Wang, B. Zhang, and Y. Tang, "DMMR: Cross-subject domain generalization for EEG-based emotion recognition via denoising mixed mutual reconstruction," in *Proc. AAAI Conf. Artif. Intell.*, 2024, vol. 38, no. 1, pp. 628–636.
- [19] D. Gao et al., "A multiscale feature fusion network based on attention mechanism for motor imagery EEG decoding," *Appl. Soft Comput.*, vol. 151, Jan. 2024, Art. no. 111129.
- [20] M. H. Rafiei, L. V. Gauthier, H. Adeli, and D. Takabi, "Self-supervised learning for electroencephalography," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 2, pp. 1457–1471, Feb. 2024.
- [21] K. Yi, Y. Wang, K. Ren, and D. Li, "Learning topology-agnostic EEG representations with geometry-aware modeling," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 36, 2023, pp. 53875–53891.
- [22] W.-B. Jiang, L.-M. Zhao, and B.-L. Lu, "Large brain model for learning generic representations with tremendous EEG data in BCI," 2024, *arXiv:2405.18765*.
- [23] Y. Zhang et al., "DMAE-EEG: A pretraining framework for EEG spatiotemporal representation learning," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 36, no. 10, pp. 17664–17678, Oct. 2025.
- [24] Y. Zhou et al., "Cross-subject cognitive workload recognition based on EEG and deep domain adaptation," *IEEE Trans. Instrum. Meas.*, vol. 72, pp. 1–12, 2023.
- [25] D. Li, S. Huang, L. Xie, Z. Wang, and J. Xu, "Neuron perception inspired EEG emotion recognition with parallel contrastive learning," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 36, no. 8, pp. 14049–14062, Aug. 2025.
- [26] W. Ye, J. Wang, L. Chen, L. Dai, Z. Sun, and Z. Liang, "Adaptive spatial-temporal aware graph learning for EEG-based emotion recognition," *Cyborg Bionic Syst.*, vol. 5, p. 0088, Jan. 2024.
- [27] G. Pfurtscheller, C. Brunner, A. Schlögl, and F. H. Lopes da Silva, "Mu rhythm (de)synchronization and EEG single-trial classification of different motor imagery tasks," *NeuroImage*, vol. 31, no. 1, pp. 153–159, May 2006.
- [28] D. J. McFarland, L. A. Miner, T. M. Vaughan, and J. R. Wolpaw, "Mu and beta rhythm topographies during motor imagery and actual movements," *Brain Topography*, vol. 12, no. 3, pp. 177–186, Mar. 2000.
- [29] W.-L. Zheng, J.-Y. Zhu, and B.-L. Lu, "Identifying stable patterns over time for emotion recognition from EEG," *IEEE Trans. Affect. Comput.*, vol. 10, no. 3, pp. 417–429, Jul. 2019.
- [30] X. Xu et al., "EEG-based mental workload assessment in a simulated flight task using single-channel brain rhythm sequence," *IEEE Trans. Instrum. Meas.*, vol. 73, pp. 1–12, 2024.
- [31] B. Liu, X. Liu, X. Jin, P. Stone, and Q. Liu, "Conflict-averse gradient descent for multi-task learning," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 34, 2021, pp. 18878–18890.
- [32] S. Koelstra et al., "Deap: A database for emotion analysis; using physiological signals," *IEEE Trans. Affect. Comput.*, vol. 3, no. 1, pp. 18–31, Jan. 2012.
- [33] C. Brunner, R. Leeb, G. Müller-Putz, A. Schlögl, and G. Pfurtscheller, "BCI competition 2008–Graz data set a," *Inst. Knowl. Discovery (Lab. Brain-Comput. Interfaces), Graz Univ. Technol.*, vol. 16, pp. 1–6, Jan. 2008.
- [34] W. L. Lim, O. Sourina, and L. P. Wang, "STEW: Simultaneous task EEG workload data set," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 26, no. 11, pp. 2106–2114, Nov. 2018.
- [35] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, 2025, pp. 5998–6008.
- [36] D. Dai et al., "Deepseekmoe: Towards ultimate expert specialization in mixture-of-experts language models," *arXiv preprint arXiv:2401.06066*, 2024.
- [37] T. Yue, L. Guo, J. Cheng, X. Gao, and J. Liu, "Ada-K routing: Boosting the efficiency of MoE-based LLMs," in *Proc. 13th Int. Conf. Learn. Represent.*, 2024, pp. 1–17.
- [38] Y. Jain, H. S. Behl, Z. Kira, and V. Vineet, "DAMEX: Dataset-aware mixture-of-experts for visual understanding of mixture-of-datasets," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 36, 2023, pp. 69625–69637.
- [39] C. Riquelme et al., "Scaling vision with sparse mixture of experts," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 34, 2021, pp. 8583–8595.
- [40] L. Yang et al., "EEG emotion recognition via identity based multi-gate mixture-of-experts network," in *Proc. IEEE Int. Conf. Bioinf. Biomed. (BIBM)*, Dec. 2022, pp. 2498–2505.
- [41] Z. Du, R. Peng, W. Liu, W. Li, and D. Wu, "Mixture of experts for EEG-based seizure subtype classification," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 31, pp. 4781–4789, 2023.
- [42] X.-H. Liu, W.-B. Jiang, W.-L. Zheng, and B.-L. Lu, "MoGE: Mixture of graph experts for cross-subject emotion recognition via decomposing EEG," in *Proc. IEEE Int. Conf. Bioinf. Biomed. (BIBM)*, Dec. 2024, pp. 3515–3520.
- [43] Y. Gui, M. Chen, Y. Su, G. Luo, and Y. Yang, "Eegmamba: Bidirectional state space model with mixture of experts for EEG multi-task classification," 2024, *arXiv:2407.20254*.
- [44] X. Wang, W. Wang, Y. Cao, C. Shen, and T. Huang, "Images speak in images: A generalist painter for in-context visual learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 6830–6839.
- [45] H. Touvron et al., "Llama: Open and efficient foundation language models," 2023, *arXiv:2302.13971*.
- [46] C. Yang, M. Westover, and J. Sun, "Biot: Biosignal transformer for cross-data learning in the wild," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 36, 2023, pp. 78240–78260.
- [47] N. Mohammadi Foumani, G. Mackellar, S. Ghane, S. Irtza, N. Nguyen, and M. Salehi, "EEG2Rep: Enhancing self-supervised EEG representation through informative masked inputs," in *Proc. 30th ACM SIGKDD Conf. Knowl. Discovery Data Mining*, Aug. 2024, pp. 5544–5555.
- [48] T. Yue et al., "BrainGPT: Unleashing the potential of EEG generalist foundation model by autoregressive pre-training," 2024, *arXiv:2410.19779*.
- [49] S. Dong, L. M. Reder, Y. Yao, Y. Liu, and F. Chen, "Individual differences in working memory capacity are reflected in different ERP and EEG patterns to task difficulty," *Brain Res.*, vol. 1616, pp. 146–156, Aug. 2015.
- [50] R. A. Jacobs, M. I. Jordan, S. J. Nowlan, and G. E. Hinton, "Adaptive mixtures of local experts," *Neural Comput.*, vol. 3, no. 1, pp. 79–87, Mar. 1991.
- [51] N. Du et al., "GLaM: Efficient scaling of language models with mixture-of-experts," in *Proc. Int. Conf. Mach. Learn.*, 2021, pp. 5547–5569.
- [52] T. Chen et al., "AdaMV-MoE: Adaptive multi-task vision mixture-of-experts," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2023, pp. 17300–17311.
- [53] N. Shazeer et al., "Outrageously large neural networks: The sparsely-gated mixture-of-experts layer," 2017, *arXiv:1701.06538*.
- [54] W. Fedus, B. Zoph, and N. Shazeer, "Switch transformers: Scaling to trillion parameter models with simple and efficient sparsity," *J. Mach. Learn. Res.*, vol. 23, no. 120, pp. 1–39, 2022.
- [55] D. Lepikhin et al., "Gshard: Scaling giant models with conditional computation and automatic sharding," 2020, *arXiv:2006.16668*.
- [56] S. Katsigiannis and N. Ramzan, "DREAMER: A database for emotion recognition through EEG and ECG signals from wireless low-cost off-the-shelf devices," *IEEE J. Biomed. Health Informat.*, vol. 22, no. 1, pp. 98–107, Jan. 2018.
- [57] W.-L. Zheng and B.-L. Lu, "Investigating critical frequency bands and channels for EEG-based emotion recognition with deep neural networks," *IEEE Trans. Auto. Mental Develop.*, vol. 7, no. 3, pp. 162–175, Sep. 2015.
- [58] M. Soleymani, J. Lichtenauer, T. Pun, and M. Pantic, "A multimodal database for affect recognition and implicit tagging," *IEEE Trans. Affect. Comput.*, vol. 3, no. 1, pp. 42–55, Jan. 2012.
- [59] G. Schalk, D. J. McFarland, T. Hinterberger, N. Birbaumer, and J. R. Wolpaw, "BCI2000: A general-purpose brain-computer interface (BCI) system," *IEEE Trans. Biomed. Eng.*, vol. 51, no. 6, pp. 1034–1043, Jun. 2004.
- [60] B. Blankertz, G. Dornhege, M. Krauledat, K.-R. Müller, and G. Curio, "The non-invasive Berlin brain-computer interface: Fast acquisition of effective performance in untrained subjects," *NeuroImage*, vol. 37, no. 2, pp. 539–550, Aug. 2007.
- [61] I. Zyma et al., "Electroencephalograms during mental arithmetic task performance," *Data*, vol. 4, no. 1, p. 14, Jan. 2019.

- [62] Z. Liang et al., “EEGFuseNet: Hybrid unsupervised deep feature characterization and fusion for high-dimensional EEG with an application to emotion recognition,” *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 29, pp. 1913–1925, 2021.
- [63] W. Li, M. Wang, J. Zhu, and A. Song, “EEG-based emotion recognition using trainable adjacency relation driven graph convolutional network,” *IEEE Trans. Cognit. Develop. Syst.*, vol. 15, no. 4, pp. 1656–1672, Dec. 2023.
- [64] Y. Zhang et al., “Unsupervised time-aware sampling network with deep reinforcement learning for EEG-based emotion recognition,” *IEEE Trans. Affect. Comput.*, vol. 15, no. 3, pp. 1090–1103, Jul. 2024.
- [65] J. Tang, Z. Ma, K. Gan, J. Zhang, and Z. Yin, “Hierarchical multimodal-fusion of physiological signals for emotion recognition with scenario adaption and contrastive alignment,” *Inf. Fusion*, vol. 103, Mar. 2024, Art. no. 102129.
- [66] V. J. Lawhern, A. J. Solon, N. R. Waytowich, S. M. Gordon, C. P. Hung, and B. J. Lance, “EEGNet: A compact convolutional neural network for EEG-based brain–computer interfaces,” *J. Neural Eng.*, vol. 15, no. 5, Oct. 2018, Art. no. 056013.
- [67] Y. Song, Q. Zheng, B. Liu, and X. Gao, “EEG conformer: Convolutional transformer for EEG decoding and visualization,” *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 31, pp. 710–719, 2023.
- [68] R. T. Schirrmeister et al., “Deep learning with convolutional neural networks for EEG decoding and visualization,” *Human Brain Mapping*, vol. 38, no. 11, pp. 5391–5420, Nov. 2017.
- [69] S. U. Amin, M. Alsulaiman, G. Muhammad, M. A. Mekhtiche, and M. S. Hossain, “Deep learning for EEG motor imagery classification based on multi-layer CNNs feature fusion,” *Future Gener. Comput. Syst.*, vol. 101, pp. 542–554, Dec. 2019.
- [70] H. Wu, S. Li, and D. Wu, “Motor imagery classification for asynchronous EEG-based brain–computer interfaces,” *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 32, pp. 527–536, 2024.
- [71] H. Banville, O. Chehab, A. Hyvärinen, D.-A. Engemann, and A. Gramfort, “Uncovering the structure of clinical EEG signals with self-supervised learning,” *J. Neural Eng.*, vol. 18, no. 4, Aug. 2021, Art. no. 046020.
- [72] X. Zhang et al., “A foundation model for lesion segmentation on brain MRI with mixture of modality experts,” *IEEE Trans. Med. Imag.*, vol. 44, no. 6, pp. 2594–2604, Jun. 2025.



**Xuange Gao** (Graduate Student Member, IEEE) received the B.S. degree from Shandong University, Jinan, China, in 2022. She is currently pursuing the Ph.D. degree with the University of Chinese Academy of Sciences, Beijing, China.

Her research interests include EEG representation learning and foundation models, affective computing, and deep learning.



**Danli Wang** received the Ph.D. degree from Beihang University, Beijing, China, in 1999.

She carried out post-doctoral research at the Institute of Automation, Chinese Academy of Sciences, Beijing, where she is a Researcher and a Professor. Her research interests include deep learning, reinforcement learning, large models, group decision-making, physiological computing, and human–computer interaction.



**Yanyan Zhao** received the Ph.D. degree from the University of Chinese Academy of Sciences, Beijing, China, in 2025.

She is currently a Post-Doctoral Researcher at Tsinghua University, Beijing. Her research interests include EEG, deep learning, emotion, decision-making, and sleep research.